

Discussion Assignment 6

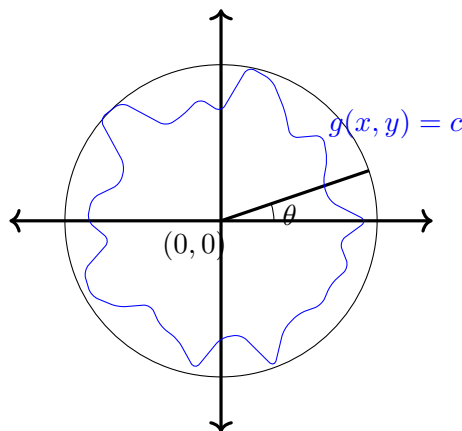
Segments of Equal Length

MATH 2210, Spring 2018

Background

The Borsuk-Ulam theorem is an example of a result in mathematics relating the topological properties of the domain of a continuous function to its range. In plain English, the Borsuk-Ulam theorem says that if our domain $\mathcal{D} \subseteq \mathbb{R}^{n+1}$ has the right *shape*, then we automatically know something about the range of any continuous function $f : \mathcal{D} \rightarrow \mathbb{R}^n$. More specifically, the Borsuk-Ulam theorem says that if S^n is the n -sphere (that is, a sphere in $n + 1$ dimensions, e.g. S^1 is the standard 2D circle, S^2 is the standard 3D sphere), then any continuous function $f : S^n \rightarrow \mathbb{R}^n$ satisfies $f(-x) = f(x)$ for at least one $x \in S^n$.

This week, we will show that this concepts holds not just for circles, but for any sufficiently 'nice' closed curve in 2 dimensions. That is, suppose that $g(x, y)$ is a continuous function, c is a real number, and the graph of $g(x, y) = c$ is a closed curve in the plane. Then, we can parameterize $g(x, y)$ by a circle – that is, suppose $g(x, y) = c$ looks something like the picture below, and note that the angle θ of the circle drawn sets points (x, y) on the graph of $g(x, y) = c$ in a direct correspondence with points (x, y) on the circle:



Of course, we could have parameterized the points (x, y) on $g(x, y) = c$ and the points on the circle quite differently had we so desired (for example, by $\theta + \pi$ instead of by θ). But in either case, we can easily construct a function which sends points on the circle to points (x, y) on $g(x, y) = c$.

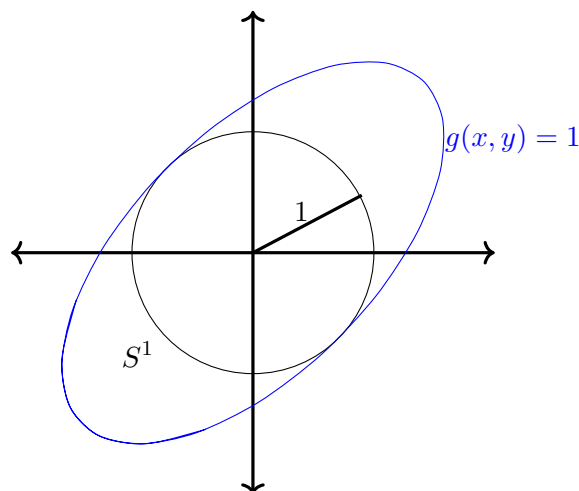
Now, let's discuss an easy fact that follows from the Borsuk-Ulam theorem. Given a circle of radius r , the point $(r, 0)$ lies at the intersection of the circle and the x -axis. So, if (x, y) is another point on the circle of radius r , then the arc length of the portion of the circle between $(r, 0)$ and (x, y) is a function which sends points on the circle to the real numbers. Let's call this function $D(x, y)$. That is, if (a, b) is a point on the circle, then $D(a, b)$ gives the length of the segment of the circle between $(r, 0)$ and (a, b) . By the Borsuk-Ulam theorem, we then know that $D(x, y) = D(-x, -y)$ for some point (x, y) on the circle. Finding this point is relatively easy – it simply corresponds to $(0, r)$. That is, if you start at $(r, 0)$ and go a quarter of the way around the circle in a counter-clockwise direction, you end up the same distance away from $(r, 0)$ as you would have if you had gone a quarter of the way around the circle in a clockwise direction.

Intuitively, if we can map the circle to another non-circular shape, then when we look at the arc length of segments of that shape, the Borsuk-Ulam theorem says that we should be able to produce the same result for the non-circular shape as well.

Suppose our function g is given by

$$g(x, y) = \frac{(x \cos(\tau) + y \sin(\tau))^2}{a^2} + \frac{(x \sin(\tau) + y \cos(\tau))^2}{b^2}$$

where τ , a , and b are constants satisfying $0 \leq \tau \leq \pi$ and $a > b > 0$. Setting $g(x, y) = 1$, we get a tilted ellipse with a major axis of length a , a minor axis of length b , and a tilt angle of τ . Visually, this amounts to the following picture



One way to parameterize the points lying on this ellipse by an angle θ (that is, one way to map the circle S^1 to the tilted ellipse) is by the function

$$r(\theta) = \langle a \cos(\theta) \cos(\tau) - b \sin(\theta) \sin(\tau), a \cos(\theta) \sin(\tau) + b \sin(\theta) \cos(\tau) \rangle$$

That is, if $\langle \cos(\theta), \sin(\theta) \rangle$ is a vector which ends at a point on the circle, then $r(\theta)$ is the vector which ends at the corresponding point on the ellipse.

If $L(x, y)$ is the function which gives the length of the shortest segment of the ellipse between $r(0)$ and $\langle x, y \rangle$ (that is, the arc length from $r(0)$ to $r(\theta)$ when $0 \leq \theta \leq \pi$ and the arc length from $r(\theta)$ to $r(0)$ when $-\pi \leq \theta \leq 0$), then the Borsuk-Ulam theorem tells us that there should exist a point (w, z) such that $L(w, z) = L(-w, -z)$. That is, there should exist an angle $\rho \in [0, \pi]$ such that

1. $r(\rho) = \langle w, z \rangle = -r(-\rho)$
2. $L(w, z) = \int_0^\rho |r'(\theta)| d\theta = \int_{-\rho}^0 |r'(\theta)| d\theta = L(-w, -z)$.

Unfortunately, solving $\int_0^\rho |r'(\theta)| d\theta$ will prove to be very difficult for any choice of a , b , and τ (go ahead, try it out). In fact, in each case, the integral $\int_0^\rho |r'(\theta)| d\theta$ turns out to be something called an *incomplete elliptic integral of the second kind*, which are of the general form

$$\int_0^\rho \sqrt{1 - k^2 \sin^2(\theta)} d\theta.$$

Elliptic integrals of the second kind can only be evaluated in terms of the so-called Jacobi elliptic functions, which are rather complicated. However, the Borsuk-Ulam theorem actually allows us to *construct identities for even these incredibly difficult integrals*. Namely, by solving 1 above, we will automatically get a new identity from 2.

This week, you will use this fact to derive an identity for incomplete elliptic integrals of the second kind. Use the Borsuk-Ulam theorem to verify one of the following identities, and select the best answer from the options below:

$$(a) \int_0^{\pi/2} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta = \int_{-\pi/2}^0 b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta$$

$$(b) \int_0^{\pi/2} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta = 2 \int_{-\pi/4}^{\pi/2} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta$$

$$(c) \int_0^{\pi/2} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta = \int_{-\pi}^{\pi} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta$$

$$(d) \int_0^{\pi/2} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta = 2 \int_{-\pi/2}^{\pi} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta$$

HINT:

$$\int_0^{\pi/2} \sqrt{a^2 \sin^2(\theta) + b^2 \cos^2(\theta)} d\theta = \int_0^{\pi/2} b \sqrt{1 - \frac{b^2 - a^2}{b^2} \sin^2(\theta)} d\theta$$

Group Work

The group work portion of your grade is worth 3 points. As a group you are allowed to submit three answers. However, incorrect answers will be counted as wrong. For example, if the correct answer is (d) and your group submits $(a), (d), (d)$ as their answers the score will be $2/3$ on this portion of the assignment.

Individual Work

Now you are to write a report, one page maximum, that convinces an audience of your peers you have correctly determined the solution to the problem above. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing mathematics properly is posted on the WyoCourses page.